Fiscal Year:	FY 2005	**Task Last Updated:**	FY 11/22/2005
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Project Title:	Mechanisms by which Selenomethionine Protects Against Space Radiation Biological Effects		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Space Biology Element:	None		
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Space Biology Special Category:	None		
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Zip Code:	19104-6072	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	02-OBPR-02
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
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This research program is focused on the mechanism(s) by which L-selenomethionine reduces the levels of biological effects associated with the exposure of cells to high-energy, highly charged particles known as HZE particles from the new NASA BAF facility at the Brookhaven National Laboratory. It is known that L-selenomethionine can reduce the levels of oxidative stress produced by cellular irradiation with x-rays, protons and HZE particles. It is expected that L-selenomethionine will serve as an effective countermeasure for other types of space radiation induced biological effects, such as the induction of cancer. The studies being performed as part of this grant are exploring the mechanisms by which L-selenomethionine can serve as a countermeasure for radiation effects, including the mechanisms by which it may regulate gene expression related to radiation induced carcinogenesis. This research project is driven by our hypothesis that radiation is capable of initiating an ongoing cellular process (by a high frequency, epigenetic event) that is characterized by a state of genetic instability; a later event or events, presumably genetic, then lead directly to

Task Description:	malignant transformation. The assumptions are that the heritable epigenetic event is a change in gene expression which is induced by the radiation exposure and that L-selenomethionine is capable of preventing, suppressing or reversing this change in gene expression. Microarray technology is being used to assess the effects of SeM on HZE particle radiation-induced changes in gene expression. These studies utilize both in vitro and in vivo model systems. Studies on malignant transformation in vitro utilize HTori-3 cells, which represent a human thyroid epithelial cell model system, and studies on gene expression and function utilize HTori-3 cells and Sprague-Dawley rats as the model systems. Studies with x-rays are also being performed so that Relative Biolgoical Effectiveness (RBE) values for the HZE particles can be determined for radiation induced malignant transformation. By studying the mechanisms by which selenomethionine protects against space radiation induced biological effects, information will be obtained relevant to the mechanisms involved in HZE particle induced radiation carcinogenesis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Knowledge of the mechanisms by which selenomethionine serves as a countermeasure for radiation induced biological effects could help to develop more effective ways to prevent radiation induced biological damage leading to the known biological effects of radiation (e.g., cell and organ toxicity, mutation and malignant transformation). Research on the mechanisms involved in the prevention of radiation induced biological effects will also have earth benefits, as life on Earth involves continual exposure to various types of radiations which are capable of producing adverse biological effects.
Task Progress:	This grant is focused on the hypothesis that L-selenomethionine (SeM) serves as a countermeasure for HZE particle radiation induced adverse biological effects through its ability to regulate the expression of genes. In a previous NASA Space Radiation Laboratory (NSRL) run at the Brookhaven National Laboratory, SeM supplemented and control HTori-3 cells were irradiated with a nontoxic (10 cGy) and a slightly toxic (20 cGy) dose of radiation from 1 GeV/n iron ions. Total RNA from these samples has been prepared and analyzed using Affymetric microarray U133A chips, and VMAxS technology (by Vialogy) and EASE analysis. For both radiation doses, SeM had a profound effect on radiation-induced changes in gene regulation. In HTori-3 cells exposed to a 10 cGy dose of radiation from iron ions, supplementation with SeM modified differential expression of 66 % of up-regulated genes (143 out of 218), and 82 % of down-regulated genes (67 out of 82). Similarly, in cells exposed to a 20 cGy dose of radiation from iron ions, supplementation with SeM modified differential expression of 70 % of up-regulated genes (123 out of 176) and 93 % of down-regulated genes (115 out of 124). These results indicate a large scale effect of SeM supplementation on regulation of gene expression in HTori-3 cells irradiated with low doses of high LET radiation from HZE particles. The HZE particle radiation-regulated gene expression which was modifiable by SeM was further analyzed using EASE over-representation analysis. EASE analysis tests for Gene Ontology classes that occur more frequently in a list of genes of interest than would be expected by chance given the prevalence of a class among all genes assayed. Results of EASE over-representation analysis indicate that in HTori-3 cells exposed to a slightly toxic, 20-cGy radiation dose from iron ions, SeM abolishes down-regulation of a significant number of genes involved in regulation of the cell cycle, including nuclear division components. A further analysis using EASE software included significantly regulated genes in HTori-3 cells irradiated in the presence of SeM that were not significantly regulated by SeM alone or by radiation alone. Genes involved in negative regulation of cell proliferation, programmed cell death, response to external stimulus and various transcriptional regulators were regulated specifically when HTori-3 cells were irradiated in the presence of SeM. Analysis of microarray data has guided us to evaluate the effects of SeM on the cell cycle in HZE particle-irradiated HTori-3 cells. SeM supplemented and control HTori-3 cells have been irradiated at NSRL as a part of the fall 2005 run and are currently being analyzed using flow cytometry. Further, due to a clustering of changes in gene expression around the transcription regulator p53, we are investigating the role of redox activation of p53 in the SeM conferred protection of HTori-3 cells against HZE particle radiation-induced malignant transformation.
Bibliography Type:	Description: (Last Updated: 08/24/2020)
Articles in Peer-reviewed Journals	Stewart, J..; Ware, J.; Fortina, P.; Breaux, J.; Gulati, S.; and Kennedy, A.R. "L-selenomethionine modulates high LET radiation-induced alterations of gene expression in cultured human thyroid cells" Oncology Reports (in press) , Jan-2006
Presentation	Stewart, J.A.; Ware, J.; Fortina, P.; Kennedy, A.R. "Differential gene expression in thyroid epithelial cells exposed to low doses of HZE particle radiation" Bioastronautics Investigators' Workshop, January 10-12, 2005 Jan-2006
Presentation	Kennedy, A.R.; Ware, J.H.; Fortina, P.;Breaux, J.K.; Gulati, S.; and Stewart, J.A. "Selenomethionine protects against space radiation biological effects" 16th Annual NASA Space Radiation Investigators' Workshop, May 15-18, 2005 Jan-2006